Data Profile & Processing

Complaint → Regulation Classifier (CMPL-REG-24)
Curated CFPB consumer-complaint narratives

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Generated: 2026-07-29 20:43 UTC
Source: CFPB Consumer Complaint Database (public)
Dataset: data/complaints/cfpb_complaints.csv - 4,000 rows
Model: CMPL-REG-24 - repo: reg-agents
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1 · Processing pipeline

stage	what it does	parameter / evidence
1 · Acquire	Two passes over the public CFPB Co	scripts/fetch_cfpb_complaints.py · NONREG_TARGE [illegible]
2 · Length filter	Keep narratives ≥ 120 chars; trunca	MIN_CHARS=120 · MAX_CHARS=1800
3 · Exact dedup	Hash of whitespace/case-normalize	0 duplicates remain (verified below)
4 · Near dedup	First-200-chars fingerprint + TF-IDF	removes template complaints that w [illegible]
5 · PII check	CFPB pre-masks PII as 'XXXX'; verifi	80% of narratives carry masking tokens
6 · Balanced sampling	Per product-issue cap to fight credit	PER_ISSUE_CAP=350 · max observed 764
7 · Weak labeling	CFPB product/issue taxonomy + key	reg_agents/common/complaints.py GULATORY
8 · Scoring holdout	5% stratified reserve carved out FIR	holdout 200 (175 / 25) h, never used in
9 · Split	80/10/10 train/val/test on the rema	train 3,040 / val 380 / test 380 xed seed

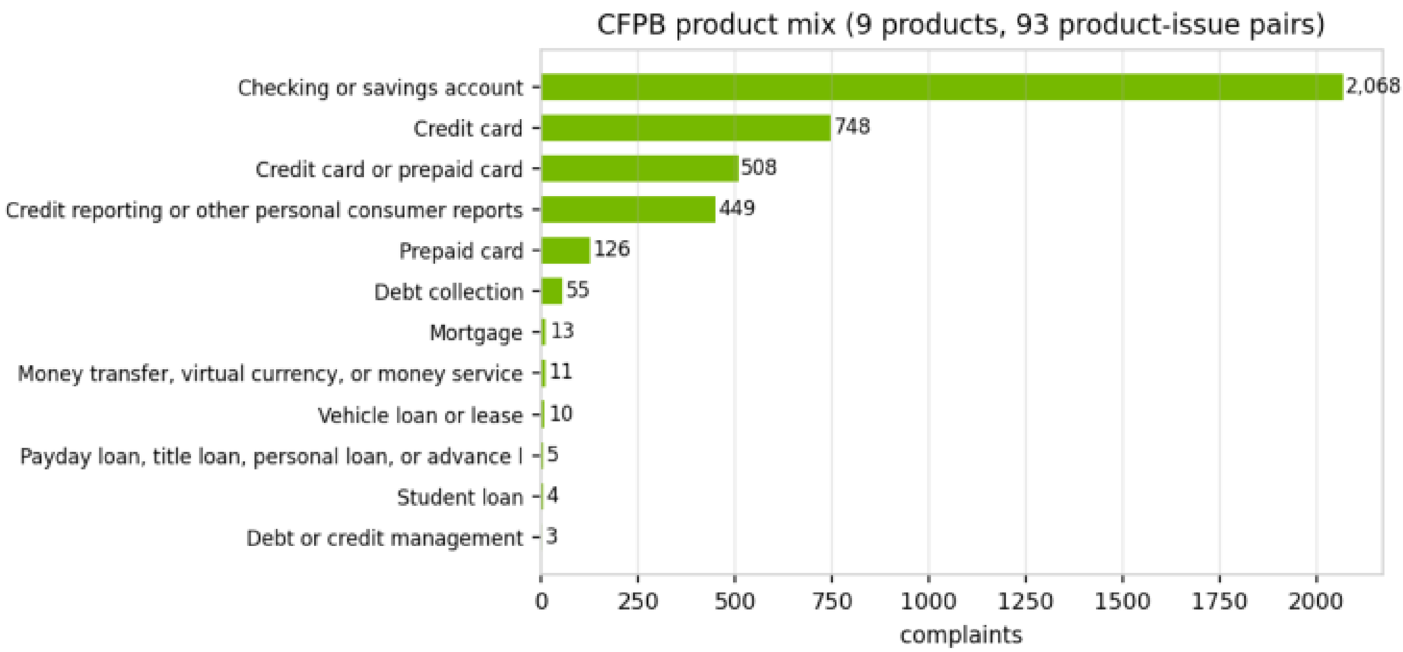
2 · Dataset schema

column	dtype	non-null	unique
complaint_id	int64	4,000	4,000
date_received	object	4,000	748
product	object	4,000	12
sub_product	object	4,000	36
issue	object	4,000	47
sub_issue	object	3,975	86
company	object	4,000	222
state	object	3,971	55
tags	object	684	3
narrative	object	4,000	4,000
label	object	4,000	24
is_regulatory	int64	4,000	2

3 · Composition

The dataset is NOT all-regulatory: 3,500 of 4,000 narratives (87.5%) carry a weak regulatory label and 500 (12.5%) are NON_REGULATORY service complaints. The raw CFPB stream is far more imbalanced (~3% non-regulatory), so acquisition runs a second, TARGETED pass over service-heavy issues ('Managing an account', 'Closing an account', ...) and enforces a non-regulatory floor at assembly time — giving the minority class enough support for stable threshold tuning and minority metrics while keeping every row a real CFPB complaint. The mix is still regulatory-dominated, which is why PR-AUC (not accuracy) is the primary stage-1 metric, why the classifiers use class weighting (class_weight='balanced'; scale_pos_weight), and why the score-distribution and calibration figures in the development document matter more than headline accuracy.

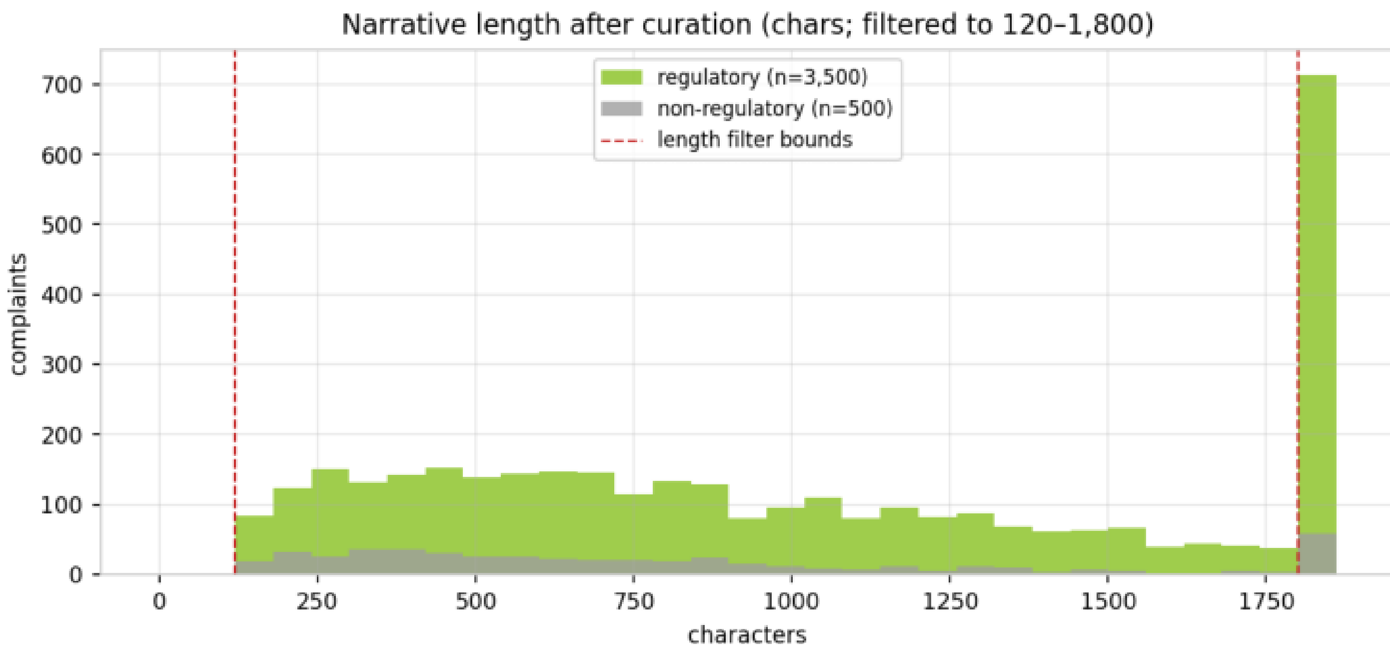
3 · Product mix



4 · Label coverage

label	category	n	share
REG_E_UNAUTHORIZED	Reg E / EFTA — Unauthoriz	1,122	28.1%
BSA_AML	BSA/AML — Account Freeze	901	22.5%
NON_REGULATORY	Non-Regulatory — General	500	12.5%
TISA_REG_DD	TISA / Reg DD — Deposit A	345	8.6%
REG_CC_FUNDS	Reg CC — Funds Availabilit	312	7.8%
FCRA_ACCURACY	FCRA — Accuracy of Repor	191	4.8%
GLBA_PRIVACY	GLBA / Privacy — Informat	121	3.0%
ECOA_DISCRIMINATION	ECOA / Reg B — Credit Dis	119	3.0%
FCRA_PERMISSIBLE_PURPO	FCRA — Permissible Purpos	114	2.9%
FCRA_INVESTIGATION	FCRA — Reinvestigation of	72	1.8%
SALES_PRACTICES	Sales Practices — Unautho	61	1.5%
FDCPA_DEBT_VALIDATION	FDCPA — Debt Validation /	42	1.1%
ECOA_ADVERSE_ACTION	ECOA / Reg B — Adverse A	25	0.6%
UDAAP	UDAAP — Unfair, Deceptiv	22	0.5%
RESPA_SERVICING	RESPA — Mortgage Servici	11	0.3%
LOAN_SERVICING	Loan Servicing (Auto/Stude	10	0.2%
FDCPA_THREATS	FDCPA — False Statements	8	0.2%
REG_Z_DISCLOSURE	Reg Z / TILA — Fees, Intere	8	0.2%
TILA_ORIGINATION	TILA — Credit Origination &	5	0.1%
REG_Z_BILLING	Reg Z / FCBA — Billing Erro	4	0.1%
FDCPA_COMMUNICATION	FDCPA — Communication T	3	0.1%
REG_E_ERROR_RESOLUTIO	Reg E / EFTA — Error Resol	2	0.1%
RESPA_LOSS_MITIGATION	RESPA — Loss Mitigation &	1	0.0%
SCRA_MLA	SCRA / MLA — Servicemen	1	0.0%

5 · Narrative length



6 · Scoring holdout + train/val/test design

Before any modeling split, a stratified 5% SCORING HOLDOUT is reserved (fixed seed). It simulates a fresh batch arriving through the ingestion layer — scripts/score_batch.py and the UI's batch-scoring upload feed it through the two-stage pipeline and emit a scored CSV (complaint id, complaint, stage-1 score, LLM reasoning). No training, validation, test, or threshold-tuning step ever touches these rows.

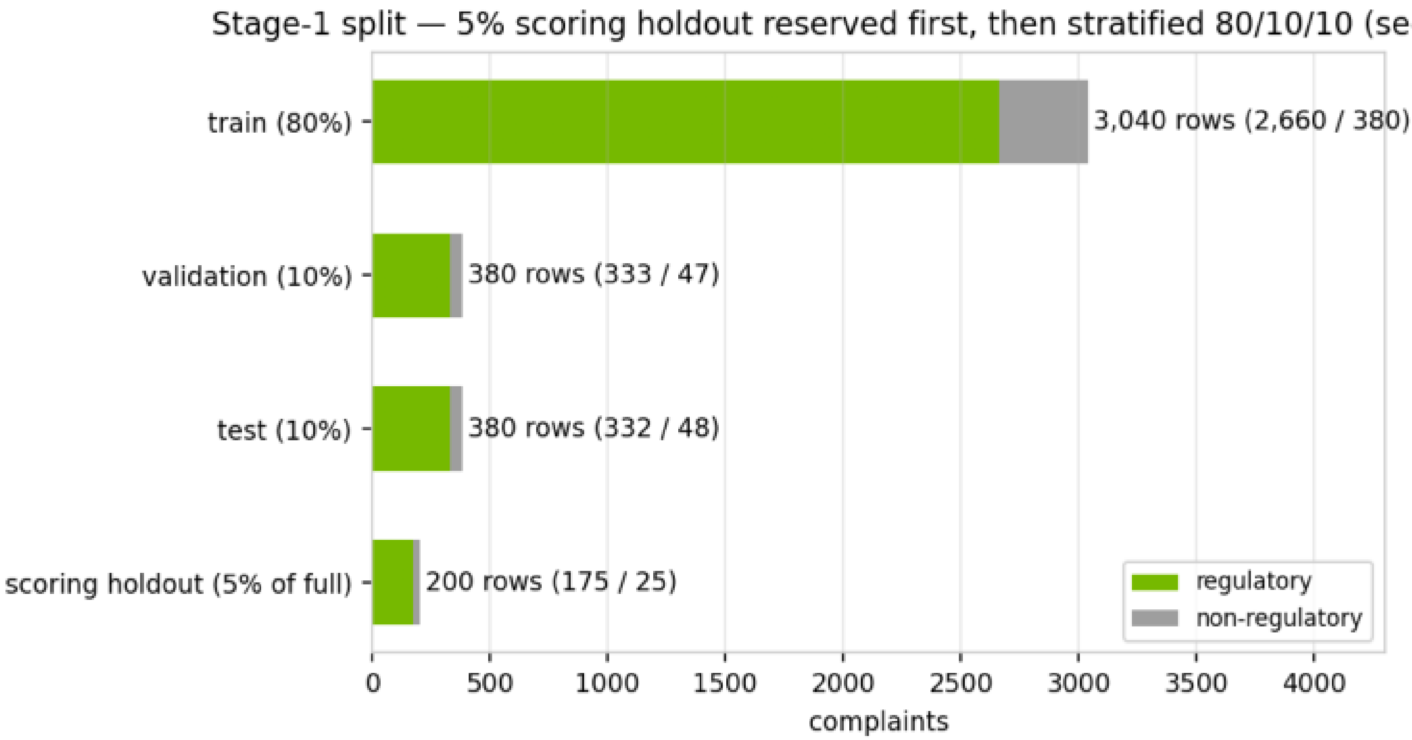
Stage 1 then uses a stratified 80/10/10 train/validation/test split of the remaining 95% (fixed seed): the test set is split off first (10%), then the remainder is split 8/1 into train and validation. Champion selection happens on validation PR-AUC only, and the decision cut-off on P(regulatory) is optimized on the same validation fold (maximizing minority-class F1) instead of assuming the default 0.5. The test set is touched exactly once, to report final metrics for the already-selected, already-thresholded model — so neither model choice nor cut-off choice can leak into reported performance.

Stage 2 involves no training: it is a prompted LLM over retrieved context. Its evaluation uses a separate stratified sample (n=115) of regulatory complaints scored against the weak labels, reported in the validation report. No complaint text is used to fit stage-2 parameters.

6 · Split composition

split	rows	regulatory	non-regulatory	reg rate
train	3,040	2,660	380	87.50%
validation	380	333	47	87.63%
test	380	332	48	87.37%
scoring holdout (5% c	200	175	25	87.50%

6 · Split (stratified)



7 · Data-quality checks (recomputed)

check	result	expectation
rows	4,000	= TARGET_ROWS (4,000)
missing values (key columns)	0	complaint_id, product, issue, narrative
exact duplicate narratives	0	post-normalization
length bounds respected	min 120 · max 1800	filter is 120–1,800 chars
PII masking present	80.2% of narratives	CFPB 'XXXX' masking tokens
max complaints per product-issue	764	cap is 350
label coverage	24 of 24 categories	all categories populated
regulatory / non-regulatory	3,500 / 500	87.5% regulatory

8 · Data update log — measured impact

Dataset acquisition is versioned, and its measured impact on the stage-1 champion (validation-tuned L1 logistic regression) is recorded here as part of data governance. **v1** used a single generic pull of the CFPB narrative stream, which left the non-regulatory minority at 135 rows (3.4%) – too few to estimate the decision cut-off or minority metrics stably, and the champion showed a persistent ~0.15 train/test ROC gap however it was regularized. **v2** added a targeted second pass over service-heavy issues with a 500-row non-regulatory floor at assembly; the improvement came from the DATA update alone – same model family, same regularization grid, same split protocol – lifting test ROC-AUC 0.849 → 0.945 and closing the generalization gap 0.147 → 0.031. **v3** (current) added a TF-IDF-cosine fuzzy-dedup pass after a leakage audit found 4.5% of test rows were near-copies (cosine ≥ 0.9) of training rows – template complaints whose openings differ, invisible to the 200-char prefix signature. Removing them makes the headline slightly LOWER but honest (ROC-AUC 0.936, gap 0.06): the v2 number was partly inflated by memorized near-duplicates. Leakage checks now run in the MDD (§9) and the test suite at every retrain.

8 · Dataset versions vs stage-1 champion

dataset version	non-reg rows	val cut-off	test ROC-AUC	train/test gap
v1 · generic pull only	135 (3.4%)	0.639	0.849	0.147
v2 · + targeted servic	500 (12.5%)	0.397	0.945	0.031
v3 · + cosine fuzzy de	500 (12.5%)	0.491	0.936	0.060
v1 → v3 net	+365 minority rows	—	+0.087	−0.087